

Hungi Jung

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- AI researcher focused on medical data modeling and clinical NLP, with hands-on experience in LLM-based EMR extraction, time-series modeling, and AWS-based deployment.

Education

 **B.S. in Computer Science, Soongsil University** (2022.03-2025.02)

Experience

Research Intern, Center for Artificial Intelligence in Healthcare, Seoul National University Bundang Hospital (Sep 2025 – Present)

- Developed an LLM-based hybrid model for gastric cancer EMR processing
- Designed a structured extraction pipeline for pathology registry variables

Undergraduate Research Intern, PEM Lab, Hongik University (Feb 2024 – Feb 2025)

- Developed an AI model for arrhythmia detection using Python and Colab
- Contributed to signal preprocessing logic and model stabilization
- Presented the research at ACK 2024 (Korea Information Processing Society)

Training

Biohealth AI Academy, AWS × Sungkyunkwan University SAY (Feb 2025 – Sep 2025)

- Participated in AWS official training on cloud computing and AI-based data analysis using Python
- Conducted pre-project and final project using AWS cloud services for biohealth AI applications

Project

Extraction and Structuring of EMR Data Using a Query-Regex-LLM Hybrid Model (Sep 2025 – Present)

- Extracted structured clinical data from SNUBH CDW to build registry-ready datasets
- Designed a hybrid pipeline combining regular expressions and LLMs for EMR parsing

- Implemented data transformation logic to standardize heterogeneous clinical records
- Developed a scalable pipeline for automated registry data generation

Machine Learning-based Mortality Prediction Dashboard using MIMIC-IV Data *(Jul 2025 – Sep 2025)*

- Extracted key features from MIMIC-IV data to predict in-hospital mortality across wards, ICUs, and emergency departments
- Trained models including XGBoost, Transformer, and Temporal Convolutional Network (TCN) for mortality prediction
- Selected the optimal model based on performance metrics and applied SHAP and LIME for model explainability
- Developed an AI-based dashboard for healthcare professionals to predict mortality rates with interpretable results

CNN + XAI-based Brain Tumor Detection *(Jun 2025 – Aug 2025)*

- Collaborated with team members to ensure model stability and reproducibility
- Utilized an open-access brain tumor dataset from Kaggle for model development
- Designed and conducted preprocessing and classification tasks using CNN models
- Applied explainable AI techniques, including Grad-CAM, to enhance model interpretability and address CNN limitations

AI-based Arrhythmia Detection Using MIT-BIH Dataset *(Jun 2024 – Apr 2025)*

- Designed and implemented a deep learning model for arrhythmia detection using ECG data from the MIT-BIH database
- Built a preprocessing pipeline incorporating cardiac domain knowledge and signal characteristics

Publications / Research

Structured LLM Pipeline for Automated Gastric Cancer Registry Construction from Bilingual Pathology Reports
(Under review, MLHC 2026 Clinical Abstract Track)

Research interests

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| • Computer Vision | • Medical AI systems & deployment (AWS) |
| • Machine Learning & Deep Learning | • Explainable AI (XAI) |
| • Clinical NLP (EMR, LLM-based extraction) | • Time-series modeling (ECG, HRV) |

Skills and Techniques

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| • Programming Languages: Python, Java, SQL | • Machine Learning & Deep Learning: PyTorch, scikit-learn, XGBoost, pandas, |
|--|---|

- Cloud Platforms: AWS
- Tools: Google Colab, Jupyter Notebook, Git
- Specialties: Machine learning, model training, classification, time-series analysis, medical signal processing, XAI

Certification

- **Network Advisor Level 2**
Information & Communication Qualification Association of Korea(ICQA).
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